

FIRST HOP REDUNDANCY PROTOCOLS (FHRP)

Comprehensive Lab Manual

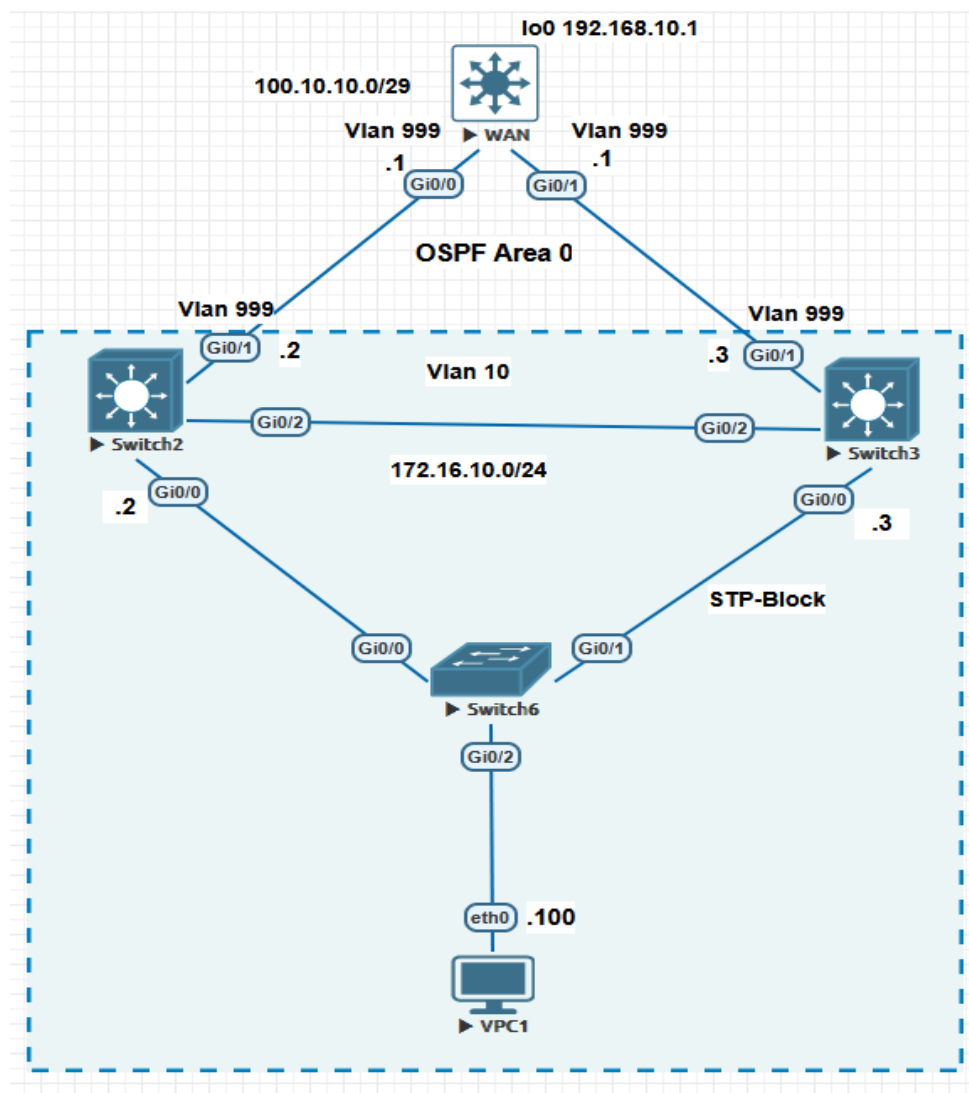


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Section 1 — Topology Overview

1.1 Network Diagram Description

The lab uses a five-device topology: one WAN router, two Layer 3 distribution switches (Switch2 and Switch3), one access switch (Switch6), and one end-host (VPC1). Three VLANs are in use: VLAN 999 (Transit/WAN), VLAN 10 (Campus/FHRP), and VLAN 99 (Management). OSPF Area 0 provides dynamic routing between the WAN router and both switches via VLAN 999. All inter-switch links are 802.1Q trunks.

1.2 VLAN Summary

VLAN ID	Name	Purpose	Subnet	Notes
999	Transit VLAN	WAN uplinks / OSPF	100.10.10.0/29	SVI on WAN, SW2 & SW3 — all OSPF peers on shared /29
10	Campus VLAN	FHRP gateway / hosts	172.16.10.0/24	SVI on SW2 & SW3; access on SW6→VPC1
99	Management	Switch management	172.16.99.0/24	SVI on SW2, SW3, SW6

1.3 Interface Address Table

Device	Interface	Connects To	Network / Role	IP Address
WAN	Gi0/0	Switch2 Gi0/1	Access port — VLAN 999	— (no IP on port)
WAN	Gi0/1	Switch3 Gi0/1	Access port — VLAN 999	— (no IP on port)
WAN	Vlan999	—	Transit SVI (OSPF active on /29 segment)	100.10.10.1 /29
Switch2	Gi0/1	WAN Gi0/0	Access port — VLAN 999 (WAN uplink)	— (no IP on port)
Switch2	Gi0/2	Switch3 Gi0/2	Trunk — VLAN 10, 99 only (inter-switch)	— (no IP on port)
Switch2	Gi0/0	Switch6 Gi0/0	Trunk — VLAN 10, 99 (to access)	— (no IP on port)
Switch2	Vlan999	—	Transit SVI (OSPF active — adjacency with WAN & SW3)	100.10.10.2 /29
Switch2	Vlan10	—	Campus SVI (OSPF passive — FHRP runs here)	172.16.10.2 /24

Device	Interface	Connects To	Network / Role	IP Address
Switch2	Vlan99	—	Management SVI (OSPF passive)	172.16.99.2 /24
Switch3	Gi0/1	WAN Gi0/1	Access port — VLAN 999 (WAN uplink)	— (no IP on port)
Switch3	Gi0/2	Switch2 Gi0/2	Trunk — VLAN 10, 99 only (inter-switch)	— (no IP on port)
Switch3	Gi0/0	Switch6 Gi0/1	Trunk — VLAN 10, 99 (STP Blocking)	— (no IP on port)
Switch3	Vlan999	—	Transit SVI (OSPF active — adjacency with WAN & SW2)	100.10.10.3 /29
Switch3	Vlan10	—	Campus SVI (OSPF passive — FHRP runs here)	172.16.10.3 /24
Switch3	Vlan99	—	Management SVI (OSPF passive)	172.16.99.3 /24
Switch6	Gi0/0	Switch2 Gi0/0	Trunk — VLAN 10, 99	— (no IP on port)
Switch6	Gi0/1	Switch3 Gi0/0	Trunk — VLAN 10, 99 (STP Blocking)	— (no IP on port)
Switch6	Gi0/2	VPC1 eth0	Access — VLAN 10	— (no IP on port)
Switch6	Vlan99	—	Management SVI	172.16.99.6 /24
VPC1	eth0	Switch6 Gi0/2	VLAN 10 host	172.16.10.100 /24

Key Addressing Points

VLAN 999 (100.10.10.0/29): All three OSPF peers share this subnet. WAN SVI = .1, SW2 SVI = .2, SW3 SVI = .3. WAN physical ports (Gi0/0, Gi0/1) are access ports in VLAN 999 — the IP lives on the Vlan999 SVI, not the physical port.

VLAN 10 (172.16.10.0/24): SW2 SVI = .2, SW3 SVI = .3. FHRP VIP = .254. VPC1 = .100.

VLAN 99 (172.16.99.0/24): SW2 = .2, SW3 = .3, SW6 = .6. Management access only — no FHRP on this VLAN.

Trunk ports carry NO IP address on the physical interface — IPs live only on the SVI (interface VlanX).

Switch2 Gi0/1 and Switch3 Gi0/1 (WAN uplinks) are ACCESS ports in VLAN 999 — the SVI Vlan999 is the routed interface for OSPF.

VLAN 999 does NOT transit between Switch2 and Switch3 — the inter-switch trunk (Gi0/2) carries only VLANs 10 and 99.

On Switch2 and Switch3: only Vlan999 SVI is OSPF active. Vlan10 and Vlan99 SVIs are OSPF passive — their subnets are advertised but no hellos are sent.

1.4 FHRP Address Assignment

A single VIP 172.16.10.254 is used for all FHRP labs on VLAN 10. Only one protocol runs at a time. VPC1 always uses 172.16.10.254 as its default gateway.

Lab	Protocol	VIP	Switch2 Role	Switch3 Role
A	HSRP v1	172.16.10.254	Active (priority 110)	Standby (priority 90)
B	HSRP v2	172.16.10.254	Active (priority 110)	Standby (priority 90)
C	VRRP v2	172.16.10.254	Master (priority 110)	Backup (priority 90)
D	VRRP v3	172.16.10.254	Master (priority 110)	Backup (priority 90)
E	GLBP	172.16.10.254	AVG + AVF1 (priority 110)	AVF2 (priority 90)

ONE PROTOCOL AT A TIME

Never run HSRP, VRRP, and GLBP simultaneously on the same interface.

Each lab section provides a CLEANUP block — always apply it before starting the next protocol.

'fhrp version vrrp v3' is a global command required for VRRP v3 — remove it before other protocols.

1.5 Routing Design

OSPF Area 0 — Transit VLAN 999

OSPF runs between WAN, Switch2, and Switch3 via VLAN 999 (100.10.10.0/29) — a shared multi-access segment.

All three OSPF peers use their Vlan999 SVI as the adjacency interface. A DR/BDR election occurs on this segment.

The WAN router sends a default route into OSPF ('default-information originate always'). Switches learn it as O*E2 — no static default needed.

VLAN 999 does NOT transit between Switch2 and Switch3. The SW2↔SW3 trunk carries only VLANs 10 and 99. SW2 and SW3 reach each other's VLAN 999 SVI only via the WAN.

On Switch2 and Switch3: Vlan999 SVI is OSPF active (non-passive) — adjacency forms here.

Vlan10 and Vlan99 SVIs on Switch2 and Switch3 are OSPF passive — their subnets are advertised into Area 0 but no hello packets are sent on those interfaces.

STP on VLAN 10/99 has no effect on OSPF adjacencies, which run on the separate VLAN 999 segment.

Section 2 — Base Configuration (Apply Before Any Lab)

Apply these configurations once. They remain unchanged across all FHRP labs. Only the FHRP-specific commands change between labs.

2.1 WAN Router

The WAN router connects to Switch2 and Switch3 via access ports (Gi0/0, Gi0/1) in VLAN 999. The IP address 100.10.10.1/29 lives on the Vlan999 SVI only — not on the physical ports. All three OSPF peers share the 100.10.10.0/29 multi-access segment. No passive-interface commands are needed on the WAN router because OSPF is enabled only on Vlan999 SVI via 'ip ospf 1 area 0'.

```
! — WAN Router —————
hostname WAN
!
! — VLAN definition —————
vlan 999
  name TRANSIT-WAN
!
! — Gi0/0: Access port — VLAN 999 (connects to Switch2 Gi0/1) —————
interface GigabitEthernet0/0
  description Access-to-Switch2-Gi0/1  [VLAN999]
  switchport mode access
  switchport access vlan 999
  no shutdown
!
! — Gi0/1: Access port — VLAN 999 (connects to Switch3 Gi0/1) —————
interface GigabitEthernet0/1
  description Access-to-Switch3-Gi0/1  [VLAN999]
  switchport mode access
  switchport access vlan 999
  no shutdown
!
! — Vlan999 SVI — this is where the IP address and OSPF live —————
interface Vlan999
  description Transit-SVI  [OSPF active — /29 multi-access with SW2 and SW3]
  ip address 100.10.10.1 255.255.255.248
  ip ospf 1 area 0
  no shutdown
!
! — ip routing and OSPF —————
ip routing
```

```

!
! No passive-interface commands needed — OSPF is enabled only on Vlan999
! via 'ip ospf 1 area 0' on the SVI. No other interface has OSPF enabled.
router ospf 1
  router-id 1.1.1.1
  default-information originate always

```

WAN Router — Same IP on Two Interfaces

The WAN router does NOT run FHRP — it is a simple upstream router.

Gi0/0 and Gi0/1 are access ports in VLAN 999. They carry no IP address — the IP 100.10.10.1/29 is on the Vlan999 SVI.

OSPF is enabled only on Vlan999 SVI via 'ip ospf 1 area 0'. No other interface has OSPF enabled, so no passive-interface commands are needed.

OSPF forms a multi-access adjacency on the 100.10.10.0/29 segment between WAN (.1), Switch2 (.2), and Switch3 (.3).

2.2 Switch2 — Base Configuration

Switch2 uses VLANs throughout. Gi0/1 (WAN uplink) is an ACCESS port in VLAN 999 — not a trunk. Gi0/2 (inter-switch to Switch3) and Gi0/0 (down to Switch6) are 802.1Q trunks carrying VLANs 10 and 99. All IP addresses live on SVIs, never on physical ports.

```

! — Switch2 Base Configuration —————
hostname Switch2

!
! — VLAN Database —————
vlan 10
  name CAMPUS
vlan 99
  name MANAGEMENT
vlan 999
  name TRANSIT-WAN
!
! — Gi0/1: ACCESS port to WAN — native VLAN 999 (not a trunk) —————
interface GigabitEthernet0/1
  description Access-to-WAN-Gi0/0  [VLAN999]
  switchport mode access
  switchport access vlan 999
  no shutdown
!
! — Gi0/2: Trunk to Switch3 — carries VLAN 10 and 99 —————

```



```
interface GigabitEthernet0/2
  description Trunk-to-Switch3-Gi0/2  [VLAN10,99]
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk allowed vlan 10,99
  no shutdown
!
! — Gi0/0: Trunk to Switch6 — carries VLAN 10 and 99 —————
interface GigabitEthernet0/0
  description Trunk-to-Switch6-Gi0/0  [VLAN10,99]
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk allowed vlan 10,99
  no shutdown
!
! — SVIs — all IP addresses live here —————
! — Vlan999: OSPF ACTIVE — adjacency with WAN (.1) and Switch3 (.3) —————
interface Vlan999
  description Transit-SVI  [OSPF active — multi-access 100.10.10.0/29]
  ip address 100.10.10.2 255.255.255.248
  ip ospf 1 area 0
  no shutdown
!
! — Vlan10: OSPF PASSIVE — subnet advertised in Area 0, no hellos sent —————
interface Vlan10
  description Campus-SVI  [FHRP runs here]  IP=172.16.10.2/24
  ip address 172.16.10.2 255.255.255.0
  ip ospf 1 area 0
  no shutdown
!
! — Vlan99: OSPF PASSIVE — subnet advertised in Area 0, no hellos sent —————
interface Vlan99
  description Management-SVI  IP=172.16.99.2/24
  ip address 172.16.99.2 255.255.255.0
  ip ospf 1 area 0
  no shutdown
!
! — Layer 3 routing —————
ip routing
!
! — OSPF — passive-interface default pattern —————
```

```
! 'passive-interface default': stops OSPF hellos on ALL interfaces.
! 'no passive-interface Vlan999': re-enables hellos on Vlan999 ONLY.
! Vlan10 and Vlan99 remain passive - subnets are advertised but no hellos.
router ospf 1
  router-id 2.2.2.2
  passive-interface default
  no passive-interface Vlan999
```

2.3 Switch3 — Base Configuration

```
! — Switch3 Base Configuration —————
hostname Switch3
!
! — VLAN Database —————
vlan 10
  name CAMPUS
vlan 99
  name MANAGEMENT
vlan 999
  name TRANSIT-WAN
!
! — Gi0/1: ACCESS port to WAN - native VLAN 999 (not a trunk) —————
interface GigabitEthernet0/1
  description Access-to-WAN-Gi0/1  [VLAN999]
  switchport mode access
  switchport access vlan 999
  no shutdown
!
! — Gi0/2: Trunk to Switch2 - carries VLAN 10 and 99 —————
interface GigabitEthernet0/2
  description Trunk-to-Switch2-Gi0/2  [VLAN10,99]
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk allowed vlan 10,99
  no shutdown
!
! — Gi0/0: Trunk to Switch6 - carries VLAN 10 and 99 —————
interface GigabitEthernet0/0
  description Trunk-to-Switch6-Gi0/1  [VLAN10,99]
  switchport trunk encapsulation dot1q
  switchport mode trunk
```

```
switchport trunk allowed vlan 10,99
no shutdown
!
! — SVIs —————
! — Vlan999: OSPF ACTIVE — adjacency with WAN (.1) and Switch2 (.2) ————
interface Vlan999
description Transit-SVI [OSPF active — multi-access 100.10.10.0/29]
ip address 100.10.10.3 255.255.255.248
ip ospf 1 area 0
no shutdown
!
! — Vlan10: OSPF PASSIVE — subnet advertised in Area 0, no hellos sent ———
interface Vlan10
description Campus-SVI [FHRP runs here] IP=172.16.10.3/24
ip address 172.16.10.3 255.255.255.0
ip ospf 1 area 0
no shutdown
!
! — Vlan99: OSPF PASSIVE — subnet advertised in Area 0, no hellos sent ———
interface Vlan99
description Management-SVI IP=172.16.99.3/24
ip address 172.16.99.3 255.255.255.0
ip ospf 1 area 0
no shutdown
!
! — Layer 3 routing —————
ip routing
!
! — OSPF — same passive-interface pattern as Switch2 —————
router ospf 1
router-id 3.3.3.3
passive-interface default
no passive-interface Vlan999
```

2.4 Switch6 — Access Switch

```
! — Switch6 Base Configuration —————
hostname Switch6
!
vlan 10
name CAMPUS
```

```
vlan 99
  name MANAGEMENT
!
! — Gi0/0: Trunk to Switch2 —————
interface GigabitEthernet0/0
  description Trunk-to-Switch2-Gi0/0  [VLAN10,99]
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk allowed vlan 10,99
  no shutdown
!
! — Gi0/1: Trunk to Switch3 (STP will block this port) —————
interface GigabitEthernet0/1
  description Trunk-to-Switch3-Gi0/0  [VLAN10,99]  [STP-BLOCK]
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk allowed vlan 10,99
  no shutdown
!
! — Gi0/2: Access port to VPC1 —————
interface GigabitEthernet0/2
  description Access-to-VPC1  [VLAN10]
  switchport mode access
  switchport access vlan 10
  spanning-tree portfast
  spanning-tree bpduguard enable
  no shutdown
!
! — Management SVI —————
interface Vlan99
  description Management-SVI  IP=172.16.99.6/24
  ip address 172.16.99.6 255.255.255.0
  no shutdown
!
! Default route for management access
ip default-gateway 172.16.99.2
```

2.5 VPC1

```
! VPC1 — VPCS syntax:
ip 172.16.10.100/24 172.16.10.254
```

```
!  
! Linux syntax:  
ip addr add 172.16.10.100/24 dev eth0  
ip route add default via 172.16.10.254
```

VPC1 Default Gateway = VIP 172.16.10.254

VPC1 always uses 172.16.10.254 as its gateway — the FHRP Virtual IP.

VPC1 never needs to know which switch is currently active.

When the active switch fails, FHRP moves the VIP to the standby switch and traffic resumes automatically.

Section 3 — OSPF Verification

OSPF must be fully converged before starting any FHRP lab. Verify adjacencies and routing table on both Switch2 and Switch3.

3.1 OSPF Verification Commands

```
! — Verify OSPF adjacencies (both switches should see WAN as neighbor) —
show ip ospf neighbor
show ip ospf neighbor detail
!
! — Verify routing table — look for O*E2 default route from WAN —
show ip route ospf
show ip route
!
! — Verify OSPF process and area —
show ip ospf
show ip ospf interface brief
show ip ospf interface Vlan999
!
! — Verify end-to-end reachability —
ping 100.10.10.1 source Vlan999
ping 172.16.10.3 source Vlan10
```

3.2 Expected OSPF State

Device	OSPF Neighbor	State	Interface
Switch2	WAN (1.1.1.1)	FULL/DR or FULL/BDR	Vlan999
Switch3	WAN (1.1.1.1)	FULL/DR or FULL/BDR	Vlan999
Switch2	Switch3 (3.3.3.3)	FULL/DR or FULL/DROTHER (multi-access)	Vlan999

OSPF Passive Interface Behaviour

Vlan999: OSPF active — hellos sent, adjacencies formed with WAN and other switch.

Vlan10: OSPF passive — subnet 172.16.10.0/24 is in the OSPF database and advertised, but no hello packets are sent. No adjacency forms on this interface.

Vlan99: OSPF passive — same as Vlan10. Subnet 172.16.99.0/24 is advertised into OSPF but no hellos are sent.

This means hosts on VLAN 10 and VLAN 99 see the subnets in the routing table, but OSPF never risks forming an unwanted adjacency with a host or rogue router on those VLANs.

OSPF Not Forming? Check These

VLAN 999 must be in the VLAN database on Switch2 and Switch3.

Switch2 Gi0/1 and Switch3 Gi0/1 must be access ports in VLAN 999 — not trunks. Confirm with 'show interfaces Gi0/1 switchport'.

SVI Vlan999 must be up/up on both switches: 'show interface Vlan999'.

WAN router interfaces must be up and in OSPF: 'show ip ospf interface brief'.

Timer mismatch: hello/dead timers must match between neighbors.

Section 4 — Spanning Tree Protocol (VLAN 10)

Three trunk links (Switch2↔Switch3, Switch2↔Switch6, Switch3↔Switch6) form a triangle loop on VLAN 10. Rapid-PVST blocks Switch6 Gi0/1 to eliminate the loop. Switch2 is forced to Root Bridge so the blocked port falls predictably on the Switch3 side of Switch6.

Why Only VLAN 10 STP?

VLAN 999 exists only on SW2 Gi0/1, SW3 Gi0/1, and WAN access ports — it does not transit between SW2 and SW3, so no loop is possible on VLAN 999.

VLAN 99 exists on Switch2↔Switch3 and both↔Switch6 — STP should also run on VLAN 99. Configure same root priorities.

VLAN 10 is the primary concern as it carries FHRP and host traffic.

4.1 STP Configuration

Switch2 — Root Bridge for VLAN 10 and 99

```
spanning-tree mode rapid-pvst
spanning-tree vlan 10 priority 24576
spanning-tree vlan 99 priority 24576
! Alternative: spanning-tree vlan 10,99 root primary
```

Switch3 — Secondary Root

```
spanning-tree mode rapid-pvst
spanning-tree vlan 10 priority 28672
spanning-tree vlan 99 priority 28672
! Alternative: spanning-tree vlan 10,99 root secondary
```

Switch6 — Default Priority (no change needed)

```
spanning-tree mode rapid-pvst
! PortFast and BPDUGuard already applied on Gi0/2 in base config
! Do NOT configure Switch6 as root
```

4.2 Expected STP Port States — VLAN 10

Switch	Interface	Connects To	STP Role	State
Switch2	Gi0/2	Switch3 Gi0/2	Designated	Forwarding
Switch2	Gi0/0	Switch6 Gi0/0	Designated	Forwarding
Switch3	Gi0/2	Switch2 Gi0/2	Root Port	Forwarding
Switch3	Gi0/0	Switch6 Gi0/1	Designated	Forwarding

Switch	Interface	Connects To	STP Role	State
Switch6	Gi0/0	Switch2 Gi0/0	Root Port	Forwarding
Switch6	Gi0/1	Switch3 Gi0/0	Alternate	BLOCKING

4.3 STP Verification

```
show spanning-tree vlan 10
show spanning-tree vlan 10 detail
show spanning-tree vlan 10 root
show spanning-tree vlan 10 blockedports
!
! Switch2: confirm 'This bridge is the root'
! Switch6: confirm Gi0/1 shows BLK / Alternate
```

Lab Task — STP Failover Test

1. Verify STP states with 'show spanning-tree vlan 10' on all three switches.
2. Shut Switch6 Gi0/0 (root port toward Switch2) — Gi0/1 should unblock.
3. Restore Gi0/0 — Gi0/1 returns to Blocking after RSTP reconvergence.
4. Confirm OSPF is NOT affected by STP changes (OSPF runs on VLAN 999 trunks).

Section 5 — Object Tracking for WAN Link Monitoring

IP SLA probes the WAN router from each switch's VLAN 999 SVI. If the probe fails (WAN is down), the track object goes Down, the FHRP priority decrements, and the standby switch becomes active.

5.1 Priority Decrement Logic

Device	Configured Priority	Decrement	Effective (WAN Down)	Result
Switch2 (Active/Master/AVG)	110	30	$110 - 30 = 80$	$80 < 90 \rightarrow$ Switch3 takes over
Switch3 (Standby/Backup/AVF)	90	20	$90 - 20 = 70$	Fallback if Switch3 WAN also fails

5.2 IP SLA + Track — Switch2

```
! — Switch2: IP SLA probes WAN router via VLAN 999 —————
ip sla 1
 icmp-echo 100.10.10.1 source-interface Vlan999
 frequency 5
ip sla schedule 1 life forever start-time now
!
track 1 ip sla 1 reachability
 delay down 10 up 15
!
! Probes WAN .1 every 5 seconds from Vlan999 SVI
! delay down 10: wait 10s before declaring track Down (avoids flap)
! delay up 15: wait 15s before declaring track Up (ensures stability)
```

5.3 IP SLA + Track — Switch3

```
! — Switch3: IP SLA probes WAN router via VLAN 999 —————
ip sla 1
 icmp-echo 100.10.10.1 source-interface Vlan999
 frequency 5
ip sla schedule 1 life forever start-time now
!
track 1 ip sla 1 reachability
 delay down 10 up 15
```

5.4 Tracking Verification

```
show ip sla statistics
show ip sla summary
show track 1
show track brief
!
! Expected: 'Reachability is Up' when OSPF/WAN is healthy
! Expected: 'Reachability is Down' after 10s when WAN fails
```

Section 6 — Lab A: HSRP Version 1

6.1 HSRP v1 Theory

HSRP v1 Key Facts

Cisco proprietary. RFC-like document: RFC 2281. Groups 0–255.
Multicast: 224.0.0.2, UDP port 1985.
Virtual MAC: 0000.0c07.acXX where XX = group number in hexadecimal.
Active router: highest priority wins (tie-break: highest real IP on SVI).
Standby router: monitors Active via hello messages — takes over if Active is silent for holdtime.
Preemption: OFF by default. Must explicitly configure 'standby X preempt'.
Timers: hello 3s / hold 10s (default). This lab uses 1s / 3s for fast convergence.
State machine: Initial → Learn → Listen → Speak → Standby → Active.

6.2 Cleanup

Always Apply Cleanup First

Run on both Switch2 and Switch3 before configuring any FHRP lab.

```
! — CLEANUP: Remove all FHRP from Vlan10 (both switches) —————
interface Vlan10
  no standby 1
  no standby 10
  no vrrp 10
  no glbp 10
!
no fhrp version vrrp v3
```

6.3 HSRP v1 Configuration

Switch2 — Active (priority 110)

```
! — Switch2: HSRP v1 on Vlan10 —————
interface Vlan10
  standby version 1
  standby 1 ip 172.16.10.254
  standby 1 priority 110
  standby 1 preempt
  standby 1 preempt delay minimum 30
  standby 1 authentication md5 key-string HSRP-LAB
  standby 1 track 1 decrement 30
```

```
standby 1 timers 1 3

!
! Group 1, priority 110, preempt ON, track decrement 30
! On WAN failure: 110 - 30 = 80 < Switch3 priority 90 → failover occurs
```

Switch3 — Standby (priority 90)

```
! — Switch3: HSRP v1 on Vlan10 —————
interface Vlan10
 standby version 1
 standby 1 ip 172.16.10.254
 standby 1 priority 90
 standby 1 preempt
 standby 1 authentication md5 key-string HSRP-LAB
 standby 1 track 1 decrement 20
 standby 1 timers 1 3
```

6.4 HSRP v1 Verification

```
show standby brief
show standby vlan 10
show standby vlan 10 all
show arp 172.16.10.254
show track 1

!
! Expected virtual MAC: 0000.0c07.ac01 (group 1 = 0x01)
! Switch2: State = Active   Switch3: State = Standby
```

6.5 Lab Tasks

Lab A-1 — Normal Operation

1. Apply base config (Sec 2), OSPF verify (Sec 3), STP (Sec 4), Tracking (Sec 5), Cleanup, then HSRP v1.
2. 'show standby brief' on both switches — confirm Active/Standby states.
3. From VPC1: 'ping 172.16.10.254 repeat 100' — all packets should succeed.
4. On VPC1 check ARP: virtual MAC 0000.0c07.ac01 mapped to 172.16.10.254.
5. From Switch2: 'show standby vlan 10' — verify hello/hold timers, VIP, and track status.

Lab A-2 — WAN Failover Test

1. Start continuous ping on VPC1: 'ping 172.16.10.254 repeat 10000'.

2. Simulate WAN failure on Switch2: shutdown Vlan999 SVI: 'interface Vlan999 / shutdown'.
(This stops IP SLA probes — simulates WAN unreachable from Switch2)
3. After 10s (delay down): track 1 goes Down on Switch2.
4. Switch2 HSRP priority decrements: $110 - 30 = 80$.
5. Switch3 priority (90) > Switch2 effective priority (80) → Switch3 becomes Active.
6. VPC1 pings may drop 1–3 packets during HSRP convergence.
7. Restore: 'no shutdown' on Switch2 Vlan999.
8. After 15s (delay up) + 30s (preempt delay): Switch2 reclaims Active.

Section 7 — Lab B: HSRP Version 2

7.1 HSRP v2 vs v1 Differences

Parameter	HSRP v1	HSRP v2
Group range	0 – 255	0 – 4095
Multicast IP	224.0.0.2	224.0.0.102
Virtual MAC	0000.0c07.acXX (XX=group hex)	0000.0c9f.fXXX (XXX=group hex)
MD5 auth	Yes	Yes
IPv6 support	No	No (use VRRP v3 for IPv6)
Group in lab	Group 1	Group 10 (matches VLAN number)

7.2 Cleanup — Remove HSRP v1

```
! — CLEANUP: Remove HSRP v1 (both switches) —————
interface Vlan10
  no standby version 1
  no standby 1
```

7.3 HSRP v2 Configuration

Switch2 — Active (priority 110)

```
! — Switch2: HSRP v2 on Vlan10 —————
interface Vlan10
  standby version 2
  standby 10 ip 172.16.10.254
  standby 10 priority 110
  standby 10 preempt
  standby 10 preempt delay minimum 30
  standby 10 authentication md5 key-string HSRP-LAB
  standby 10 track 1 decrement 30
  standby 10 timers 1 3
!
! Group 10 matches VLAN 10 – best practice for operational clarity
! Virtual MAC: 0000.0c9f.f00a (group 10 = 0x00a)
```

Switch3 — Standby (priority 90)

```
! — Switch3: HSRP v2 on Vlan10 —————
interface Vlan10
  standby version 2
```

```
standby 10 ip 172.16.10.254
standby 10 priority 90
standby 10 preempt
standby 10 authentication md5 key-string HSRP-LAB
standby 10 track 1 decrement 20
standby 10 timers 1 3
```

7.4 Verification

```
show standby brief
show standby vlan 10
show arp 172.16.10.254
! Expected virtual MAC: 0000.0c9f.f00a (group 10 = 0x00a)
```

Lab B-1 — HSRP v2 Failover (same steps as Lab A-2)

1. Apply cleanup, then HSRP v2 config above on both switches.
2. Verify Active/Standby states and new virtual MAC format.
3. Shut Switch2 Vlan999 → track down → Switch2 priority 80 → Switch3 Active.
4. Note: vMAC changes from HSRP v1 format (0000.0c07.ac01) to v2 (0000.0c9f.f00a).
5. Restore Switch2 Vlan999 → preempt back after delay timers.

Section 8 — Lab C: VRRP Version 2

8.1 VRRP v2 Theory

VRRP v2 Key Facts (RFC 3768)

Open standard (interoperable). Groups 1–255. IP protocol 112, Multicast 224.0.0.18.
 Virtual MAC: 0000.5e00.01XX where XX = group number in hexadecimal.
 Master: highest priority (default 100). The IP Owner (real IP = VIP) always wins at priority 255.
 VIP 172.16.10.254 is NOT owned by either switch (.2 or .3) — priority determines Master.
 Preemption: ON by default (unlike HSRP — no explicit 'preempt' needed).
 IPv4 only — no IPv6 support. Use VRRP v3 for IPv6.
 No built-in MD5 authentication in standard (deprecated). Not configured in this lab.
 Advertisement interval default: 1s. Master Down interval: 3 × adv interval + skew time.

8.2 Cleanup — Remove HSRP v2

```
! — CLEANUP: Remove HSRP v2 (both switches) —————
interface Vlan10
  no standby version 2
  no standby 10
```

8.3 VRRP v2 Configuration

Switch2 — Master (priority 110)

```
! — Switch2: VRRP v2 on Vlan10 —————
interface Vlan10
  vrrp 10 ip 172.16.10.254
  vrrp 10 priority 110
  vrrp 10 preempt
  vrrp 10 preempt delay minimum 30
  vrrp 10 track 1 decrement 30
  vrrp 10 timers advertise 1
  vrrp 10 timers learn
!
! Virtual MAC: 0000.5e00.010a (group 10 = 0x0a)
```

Switch3 — Backup (priority 90)

```
! — Switch3: VRRP v2 on Vlan10 —————
interface Vlan10
  vrrp 10 ip 172.16.10.254
```

```
vrrp 10 priority 90
vrrp 10 preempt
vrrp 10 track 1 decrement 20
vrrp 10 timers advertise 1
vrrp 10 timers learn
```

8.4 Verification

```
show vrrp brief
show vrrp interface vlan 10
show vrrp all
show arp 172.16.10.254
! Expected virtual MAC: 0000.5e00.010a (group 10 = 0x0a)
! Switch2: State = Master   Switch3: State = Backup
```

Lab C-1 — VRRP v2 Failover

1. Apply cleanup, then VRRP v2 config above.
2. 'show vrrp brief' → Switch2: Master, Switch3: Backup.
3. Shut Switch2 Vlan999 → IP SLA fails → track Down → Switch2 priority 80 → Switch3 Master.
4. Wireshark tip: filter 'ip.dst == 224.0.0.18' to capture VRRP advertisements.
5. Restore Switch2 Vlan999 → Switch2 preempts back as Master after delay.

Section 9 — Lab D: VRRP Version 3

9.1 VRRP v3 Theory

VRRP v3 Key Facts (RFC 5798)

Open standard. Superset of VRRP v2 — adds full IPv6 support and centisecond timers.
 IOS-XE global command required: 'fhrp version vrrp v3' (enables v3 for all interfaces).
 Interface syntax changes: 'vrrp X address-family ipv4' block replaces 'vrrp X ip'.
 Virtual MAC: same format as v2: 0000.5e00.01XX.
 No authentication in v3 (removed from RFC 5798).
 Preemption ON by default. Object tracking uses 'track X decrement Y' inside address-family block.
 Timers in centiseconds: 100 cs = 1 second. Allows sub-second tuning.
 VRRP v2 and v3 CANNOT coexist on same interface group — remove v2 before configuring v3.

9.2 Cleanup — Remove VRRP v2

```
! — CLEANUP: Remove VRRP v2 (both switches) —————
interface Vlan10
  no vrrp 10
!
! Also ensure no stale global vrrp v3 command (in case of prior lab):
no fhrp version vrrp v3
```

9.3 VRRP v3 Configuration

Switch2 — Master (priority 110)

```
! — Switch2: VRRP v3 on Vlan10 —————
fhrp version vrrp v3
!
interface Vlan10
  vrrp 10 address-family ipv4
    address 172.16.10.254 primary
    priority 110
    preempt delay minimum 30
    track 1 decrement 30
    timers 100 300
  !
!
! timers 100 300 = advertise every 100cs (1s), master-down at 300cs (3s)
! Virtual MAC: 0000.5e00.010a (same as VRRP v2, group 10 = 0x0a)
```

Switch3 — Backup (priority 90)

```
! — Switch3: VRRP v3 on Vlan10 —  
fhrp version vrrp v3  
!  
interface Vlan10  
  vrrp 10 address-family ipv4  
    address 172.16.10.254 primary  
    priority 90  
    preempt delay minimum 15  
    track 1 decrement 20  
    timers 100 300  
!
```

9.4 Verification

```
show vrrp brief  
show vrrp vlan 10 ipv4  
show vrrp all  
show arp 172.16.10.254
```

Lab D-1 — VRRP v3 Failover

1. Apply cleanup ('no vrrp 10' + 'no fhrp version vrrp v3'), then VRRP v3 config.
2. 'show vrrp brief' → Switch2: Master, Switch3: Backup.
3. Shut Switch2 Vlan999 → track Down → Switch2 priority 80 → Switch3 becomes Master.
4. Compare with VRRP v2 — centisecond timers allow finer tuning (e.g., 50cs hello = 500ms).
5. Restore Switch2 Vlan999 → preempts back after delay.

Section 10 — Lab E: GLBP

10.1 GLBP Theory

GLBP Key Facts

Cisco proprietary. Groups 1–1024. Multicast 224.0.0.102, UDP port 3222.

AVG (Active Virtual Gateway): elected by highest priority. Handles all ARP requests for the VIP and assigns different virtual MACs to each client in rotation.

AVF (Active Virtual Forwarder): each router including the AVG acts as an AVF. It actually forwards packets for clients that learned its virtual MAC.

Virtual MAC format: 0007.b400.XXYY — XX = group number in hex, YY = forwarder number (01, 02...).

Load balancing: round-robin (default) means each new ARP gets the next forwarder's MAC — different clients use different physical gateways simultaneously.

GLBP is the only Cisco FHRP providing active/active load balancing.

OBJECT TRACKING in GLBP uses the WEIGHTING mechanism — there is no 'glbp X track Y decrement Z' command.

Correct syntax: 'glbp X weighting <max> lower <L> upper <U>' sets thresholds, then 'glbp X weighting track Y decrement Z' ties a track object to the weight.

When the tracked object goes Down the weight decrements. If weight reaches or drops below the lower threshold, the AVF stops forwarding (moves to Listen state).

The upper threshold controls recovery — the AVF resumes forwarding only when the weight rises back above the upper value.

AVG election is still controlled by 'glbp X priority' and preempt — weighting only controls AVF forwarding state.

10.2 GLBP Virtual MAC Assignment in This Lab

Device	GLBP Role	Forwarder	Virtual MAC / Weighting
Switch2	AVG + AVF1	Forwarder 1	0007.b400.0a01 weight 110, lower 80, upper 100
Switch3	AVF2	Forwarder 2	0007.b400.0a02 weight 90, lower 60, upper 80

10.3 Cleanup — Remove VRRP v3

```
! — CLEANUP: Remove VRRP v3 (both switches) —————
interface Vlan10
  no vrrp 10
!
no fhrp version vrrp v3
```

10.4 GLBP Configuration

Switch2 — AVG (priority 110)

```
! — Switch2: GLBP on Vlan10 —————
interface Vlan10
  glbp 10 ip 172.16.10.254
  glbp 10 priority 110
  glbp 10 preempt
  glbp 10 preempt delay minimum 30
  glbp 10 authentication md5 key-string GLBP-LAB
  glbp 10 timers 1 3
  glbp 10 load-balancing round-robin
  glbp 10 forwarder preempt
  glbp 10 forwarder preempt delay minimum 30
!
! GLBP tracking uses the weighting mechanism – not 'glbp X track' directly
! Step 1: set weighting thresholds (max weight, lower and upper bounce values)
  glbp 10 weighting 110 lower 80 upper 100
!
! Step 2: tie track object to weighting – decrement weight when track goes Down
  glbp 10 weighting track 1 decrement 30
!
! When track 1 goes Down: weight drops from 110 to 80 (= lower threshold)
! At or below lower threshold the AVF stops forwarding (goes to Listen state)
! AVG role is governed by 'glbp 10 priority' – preempt handles AVG failover
! AVG role: Switch2 wins priority 110 vs Switch3 priority 90
! AVG distributes ARP replies: client A → vMAC 0a01, client B → vMAC 0a02, etc.
```

Switch3 — AVF2 (priority 90)

```
! — Switch3: GLBP on Vlan10 —————
interface Vlan10
  glbp 10 ip 172.16.10.254
  glbp 10 priority 90
  glbp 10 preempt
  glbp 10 authentication md5 key-string GLBP-LAB
  glbp 10 timers 1 3
  glbp 10 load-balancing round-robin
  glbp 10 forwarder preempt
  glbp 10 forwarder preempt delay minimum 30
!
  glbp 10 weighting 90 lower 60 upper 80
  glbp 10 weighting track 1 decrement 30
```

10.5 Verification

```

show glbp brief
show glbp vlan 10
show glbp vlan 10 detail
show arp 172.16.10.254
show glbp vlan 10 | include Forwarder
show glbp vlan 10 | include weight
!
! Check weighting and track state:
show glbp vlan 10 | include weight
show track 1
!
! Expected when healthy:
!   Weighting 110   lower 80   upper 100   - tracking is Up
! Expected when track 1 Down:
!   Weighting 80    lower 80   upper 100   - at lower threshold, AVF goes Listen

```

Lab E-1 — GLBP Load Balancing Verification

1. Apply cleanup, then GLBP config above on both switches.
2. 'show glbp brief' → Switch2: Active/AVG, Switch3: Listen/AVF.
3. From VPC1: clear ARP 'arp -d *' then ping multiple hosts in VLAN 10.
4. Watch ARP replies for 172.16.10.254 — they alternate between 0007.b400.0a01 and 0a02.
5. 'show glbp vlan 10 detail' → check 'Pkts redirected' counters and weighting values on each forwarder.

Lab E-2 — GLBP AVG Failover

1. Shut Switch2 Vlan999 → IP SLA fails → track 1 goes Down.
2. Switch2 weighting decrements: $110 - 30 = 80$ (reaches lower threshold of 80).
3. At or below lower threshold Switch2 AVF moves to Listen — stops forwarding.
4. Switch3 becomes the sole active forwarder for all VLAN 10 traffic.
5. AVG role: Switch2 priority (110) still beats Switch3 (90) so Switch2 remains AVG unless it also loses GLBP hellos. If Switch2 goes fully down, Switch3 wins AVG via priority.
6. Restore Switch2 Vlan999 → track 1 recovers → weight rises.
7. Weight must exceed upper threshold (100) before Switch2 AVF resumes — prevents flapping.
8. 'debug glbp events' to observe weighting and AVF state transitions (lab use only).

Section 11 — Protocol Comparison and Troubleshooting

11.1 Side-by-Side Protocol Comparison

Feature	HSRP v1	HSRP v2	VRRP v3	GLBP
Standard	Cisco	Cisco	RFC 5798	Cisco
VIP (this lab)	172.16.10.254	172.16.10.254	172.16.10.254	172.16.10.254
Group used	1	10	10	10
Multicast	224.0.0.2	224.0.0.102	224.0.0.18	224.0.0.102
Virtual MAC	0000.0c07.ac01	0000.0c9f.f00a	0000.5e00.010a	0007.b400.0a01/02
Active role name	Active	Active	Master	AVG + AVF
Preempt default	OFF	OFF	ON	ON
Load balancing	No	No	No	Yes (round-robin AVF)
Auth	MD5	MD5	None (v3)	MD5
Global command	None	None	fhrp version vrrp v3	None
Track mechanism	standby X track Y decrement Z	standby X track Y decrement Z	track Y decrement Z (under addr-family)	glbp X weighting track Y decrement Z

11.2 Common Issues and Fixes

Symptom	Likely Cause	Fix
Both switches show Active/Master	Version or group mismatch; auth mismatch	Match version, group, and auth key on both switches
FHRP failover does not occur	Decrement too small; preempt not set	priority – decrement must be < other switch priority; add preempt
VPC1 cannot reach 172.16.10.254	Vlan10 SVI down or FHRP misconfigured	'show interface Vlan10' must show up/up; check VIP config
VRRP v3 syntax rejected	VRRP v2 active or no global command	Remove 'vrrp X ip' first; add 'fhrp version vrrp v3'
IP SLA stays Down after WAN restore	delay up 15s timer still running	Wait 15s; check 'show track 1' for time-to-change
OSPF neighbor not forming	Vlan999 SVI down or VLAN 999 not allowed on trunk	Check 'show interfaces trunk'; 'show interface Vlan999'
VLAN 10 SVI is down/down	VLAN not in database or trunk not allowing it	'show vlan brief' — VLAN 10 must be active on all switches
GLBP forwarder not forwarding	Weight below threshold or track object down	'show glbp vlan 10 detail'; 'show track 1'
STP blocking unexpected port	Priority not set on Switch2	Apply 'spanning-tree vlan 10 priority 24576' on Switch2

11.3 Debug Commands

Lab Use Only — High CPU Impact

Always run 'undebg all' when finished to stop debugging.

```
! — HSRP Debug —————
debug standby events
debug standby terse
!
! — VRRP Debug —————
debug vrrp all
debug vrrp events
debug vrrp packets
!
! — GLBP Debug —————
debug glbp all
debug glbp events
!
! — Object Tracking / IP SLA —————
debug track
debug ip sla trace
!
! — OSPF —————
debug ip ospf adj
debug ip ospf hello
!
! — Stop all debugging —————
undebg all
```

Section 12 — Quick Reference

12.1 Complete IP Address Reference

Device	Interface	Description	Subnet	IP Address
WAN	Gi0/0	To Switch2 — OSPF	100.10.10.0/29	100.10.10.1
WAN	Gi0/1	To Switch3 — OSPF	100.10.10.0/29	100.10.10.1
Switch2	Vlan999	Transit SVI — OSPF	100.10.10.0/29	100.10.10.2
Switch2	Vlan10	Campus SVI — FHRP	172.16.10.0/24	172.16.10.2
Switch2	Vlan99	Management SVI	172.16.99.0/24	172.16.99.2
Switch3	Vlan999	Transit SVI — OSPF	100.10.10.0/29	100.10.10.3
Switch3	Vlan10	Campus SVI — FHRP	172.16.10.0/24	172.16.10.3
Switch3	Vlan99	Management SVI	172.16.99.0/24	172.16.99.3
Switch6	Vlan99	Management SVI	172.16.99.0/24	172.16.99.6
VPC1	eth0	VLAN 10 host	172.16.10.0/24	172.16.10.100
FHRP VIP	Vlan10	Shared virtual gateway	172.16.10.0/24	172.16.10.254

12.2 Show Commands by Protocol

Protocol	Primary Show Commands	State Keywords
HSRP v1/v2	show standby brief / show standby vlan 10	Active / Standby / Init
VRRP v2	show vrrp brief / show vrrp all	Master / Backup / Init
VRRP v3	show vrrp vlan 10 ipv4 / show vrrp brief	Master / Backup / Init
GLBP	show glbp brief / show glbp vlan 10 detail	Active/Listen + AVG/AVF
Tracking	show track 1 / show ip sla statistics	Up / Down + last change
OSPF	show ip ospf neighbor / show ip route ospf	FULL / 2WAY / EXSTART
STP	show spanning-tree vlan 10	FWD / BLK / LRN + role
Trunks	show interfaces trunk / show vlan brief	VLANs in active list

12.3 Lab Sequence Checklist

Follow this order for every FHRP lab:

1. Apply base configuration — Section 2 (all devices). Do once only.
2. Verify OSPF — Section 3. SW2 and SW3 must have FULL adjacency with WAN.
3. Configure STP — Section 4. Confirm Switch2 is root, Switch6 Gi0/1 is Blocking.
4. Configure Object Tracking — Section 5. Confirm 'show track 1' shows Up.
5. Apply CLEANUP block — remove any previous FHRP from Vlan10.
6. Apply target FHRP protocol — Sections 6–10.
7. Verify with show commands — confirm Active/Master/AVG roles.
8. Run failover test — shut Switch2 Vlan999, observe FHRP transition, restore.

9. Apply CLEANUP before switching to the next protocol.

12.4 Port Mode and VLAN Reference

Device	Interface	Connects To	Port Mode / VLANs
WAN	Gi0/0	Switch2 Gi0/1	Access — VLAN 999
WAN	Gi0/1	Switch3 Gi0/1	Access — VLAN 999
WAN	Vlan999	—	SVI — OSPF active (100.10.10.1/29)
Switch2	Gi0/1	WAN Gi0/0	Access — VLAN 999 (WAN uplink)
Switch2	Gi0/2	Switch3 Gi0/2	Trunk — VLAN 10, 99 only (no VLAN 999)
Switch2	Gi0/0	Switch6 Gi0/0	Trunk — VLAN 10, 99
Switch2	Vlan999	—	SVI — OSPF active (100.10.10.2/29)
Switch2	Vlan10	—	SVI — OSPF passive (172.16.10.2/24)
Switch2	Vlan99	—	SVI — OSPF passive (172.16.99.2/24)
Switch3	Gi0/1	WAN Gi0/1	Access — VLAN 999 (WAN uplink)
Switch3	Gi0/2	Switch2 Gi0/2	Trunk — VLAN 10, 99 only (no VLAN 999)
Switch3	Gi0/0	Switch6 Gi0/1	Trunk — VLAN 10, 99 (STP Blocking)
Switch3	Vlan999	—	SVI — OSPF active (100.10.10.3/29)
Switch3	Vlan10	—	SVI — OSPF passive (172.16.10.3/24)
Switch3	Vlan99	—	SVI — OSPF passive (172.16.99.3/24)
Switch6	Gi0/0	Switch2 Gi0/0	Trunk — VLAN 10, 99
Switch6	Gi0/1	Switch3 Gi0/0	Trunk — VLAN 10, 99 (STP Blocking)

Final Verification Tip

Before any FHRP test: 'show interfaces trunk' on all three switches — VLAN 10 must appear in all three columns (allowed, active, forwarding).

If Vlan10 SVI is down/down: check 'show vlan brief' — VLAN 10 must be active (not suspended or inactive).

OSPF must be FULL before FHRP tracking works — IP SLA probes WAN via Vlan999.

After FHRP failover: check 'show arp' on VPC1 — the virtual MAC must remain 172.16.10.254's MAC (HSRP/VRRP) or change to the other forwarder (GLBP).

— End of Lab Manual —